CODE SPORT

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In this month's column, we will look at how software upgrades could lead to failure in some cases, and how online software updates can help to reduce application downtime.

In the last couple of months, we looked at various memory errors that lead to application crashes and/or decreased performance, and discussed a number of tools that help detect these errors. For enterprise applications, 100 per cent availability remains the goal, and any application crash that causes down-time or decreased performance needs to be prevented at all costs. Let us assume that we have attained the ultimate wisdom regarding programming, and that we have developed a software application that can be certified as having zero bugs. Now, while selling this software to Company X, the firm wants an assurance that deploying our certifiable bug-free software in place of its current ancient, bug-ridden software application will not cause any issues. Now, would you make that guarantee to Company X?

Well, the answer is that you cannot make that guarantee. All that you know is that your application is bug-free. But remember the important fact that your application is getting deployed in an existing software environment, where it has to interact with multiple other applications. It needs to be configured correctly, and adequate resources should be made available for its correct operation. All these factors are not in your control as a software developer. Therefore, even if we can certify that our software is 100 per cent bug-free, it cannot be assumed that deploying it in a enterprise environment will be trouble-free. In fact, a software engineering study found that the major root cause of service disruptions that occur when software application upgrades are performed in enterprise software environments was the bugs in the upgrade process itself. These include broken dependencies between different pieces of software components, failures due to configuration, file errors, etc. Therefore, software upgrade failures are a major contributor to application crashes, and need to be studied and understood well in advance in order to avoid them.

In this month's column, we will focus our attention on software upgrade failures, discussing the types of failures that occur during the upgrade process, and what kind of upgrade process can help reduce such occurrences. We will then briefly cover online software upgrades that ensure zero downtime for the end customer. In fact, it is not just user applications that can be upgraded online—even operating-system kernels can be patched and fixed for critical defects or security issues, without having to reboot the system in order to apply the upgrade. Let's look at Ksplice, a mechanism for online software updates for the Linux kernel.

Software upgrade failures

In the summer of 1996, America Online (*http://www.aol.com/*) was one of the major Internet service providers that advertised its high reliability, with claims to being immune to network outages. However, at 4 a.m. EDT (Eastern Daylight Time) on 7 Aug